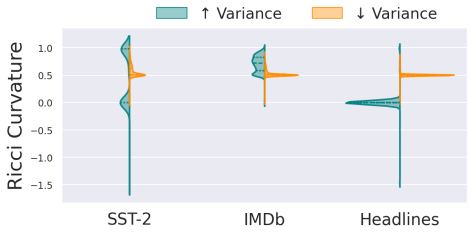


Figure 9: Final layer ORC distributions in BERT-Tiny trained via regularization methods increasing (↑) and decreasing (↓) variance, exhibiting a connection with the direction of variance regularization.

**Manipulating curvature distribution of language transformers.** The attention curvature distributions obtained through our proposed variance regularization applied to BERT-Tiny trained from initialized weights echo the same results as observed for vision models. Regularizing the loss to decrease the attention score variance discourages smaller curvature values as expected (Figure 9). This is observed for BERT-Tiny in two out of three datasets. Note that training language models from initialized weights requires substantially more computational resources than training L-ViT, and as a result, the performance does not reach the level of accuracy obtained by fine-tuning pre-trained language transformers. Furthermore, since the attention graph is substantially larger and curvature computation more expensive, we use a smaller sample size, leading to larger variance across data and trials.

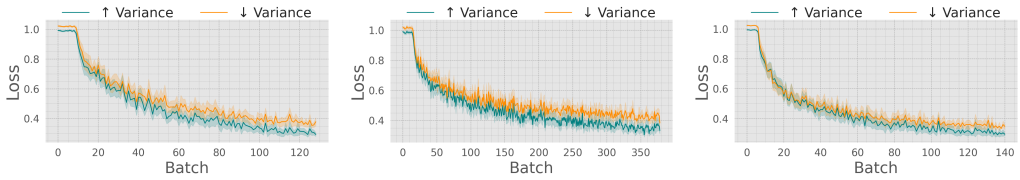


Figure 10: Loss minimization on BERT-Tiny with regularization methods increasing (↑) and decreasing (↓) variance for Headlines (left), SST-2 (middle), and IMDb (right). Higher variance can promote faster loss reduction and achieve convergence in fewer iterations.

**Loss minimization in variance-regularized training of language transformers.** We also observe larger loss variance across trials in language models compared to vision models, and due to computational constraints, analyzing and reducing this variance is beyond the scope of this project, especially considering strong empirical validations already observed using *L-ViT*. As a result, the loss dynamics for training BERT-Tiny with variance regularization, shown in Figure 10, does not (yet) yield statistically significant differences between increasing and decreasing variance. Having said that, the average over 30 trials on BERT-Tiny is consistent with what we observe for *L-ViT* in Section 4.2, though these average values fall within the 95% confidence interval of each other.

### E.2 Forgetfulness and Curvature Distribution

As a part of our empirical analysis of the robustness properties of transformers, in Section 5 we investigated the connection between ORC distribution and *forgetfulness* of pre-trained transformers. Recall that for this experiment, we first fine-tune each model on a primary dataset, and subsequently fine-tune it on a secondary dataset, monitoring how performance on test sets on both datasets change throughout the second round of fine-tuning. A fast decay in performance on the primary dataset implies a more *forgetful* model. As discussed in Section 5, our results indicate that DeiT and BERT-Tiny tend to forget learned patterns faster than ViT and ELECTRA-Small, signaling higher robustness of ViT and ELECTRA-Small to training on new distributions.

Meanwhile, as shown in Figure 6, the ORC distributions of attention maps in ViT and ELECTRA-Small lean further towards more positive ORC values. Together, these observations support our arguments regarding the positive correlation between the ORC of attention maps and the robustness

of transformers, as the more positive-leaning distribution belongs to the less forgetful model. The forgetfulness observations are demonstrated in Figure 7 for CIFAR10 and Headlines as the primary datasets, and MNIST and SST-2 as secondary datasets. We now include results on the other pairs of primary and secondary datasets. As Figure 11 demonstrates these results for vision transformers, the same trend as in Figure 7 is observed on most other pairs of datasets, further validating our results from this experiment. Figure 12 shows the results on other pairs for language transformers, where the same general trend is observed when forgetting occurs. It is worth noting that the forgetting phenomenon is not observed in some of our experiments on language models, which is likely due to semantic similarities between the tasks and datasets in hand.

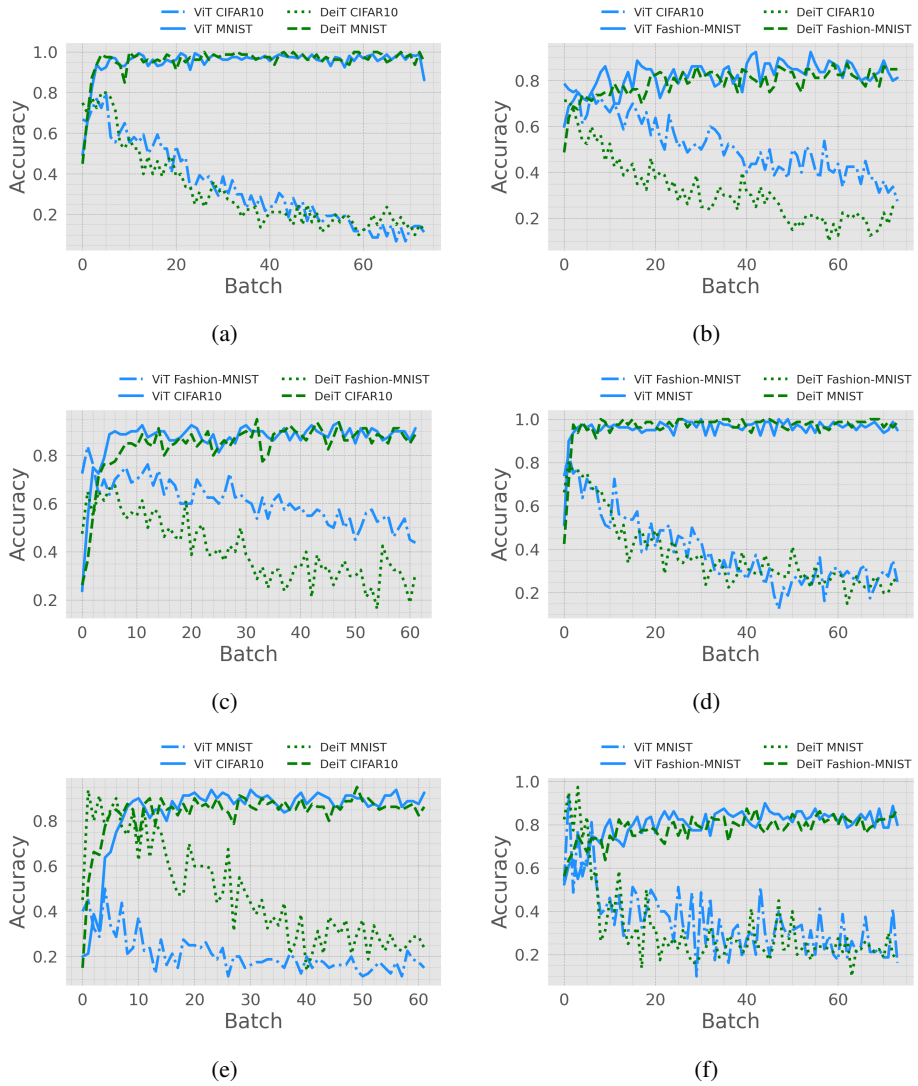


Figure 11: Test set accuracy of ViT (blue) and DeiT (green) models while fine-tuning on a secondary dataset after fine-tuning on a primary dataset. The primary and secondary datasets are: (a) CIFAR10 and MNIST; (b) CIFAR10 and Fashion-MNIST; (c) Fashion-MNIST and CIFAR10; (d) Fashion-MNIST and MNIST; (e) MNIST and CIFAR10; (f) MNIST and Fashion-MNIST. DeiT demonstrates a faster degradation in performance compared to ViT following subsequent fine-tuning.

### E.3 Training Robust Transformers

Given the established success of connection-dropping methods in training robust neural networks (Goodfellow et al., 2014; Jain et al., 2015; Srivastava et al., 2014; Yang et al., 2020), we also

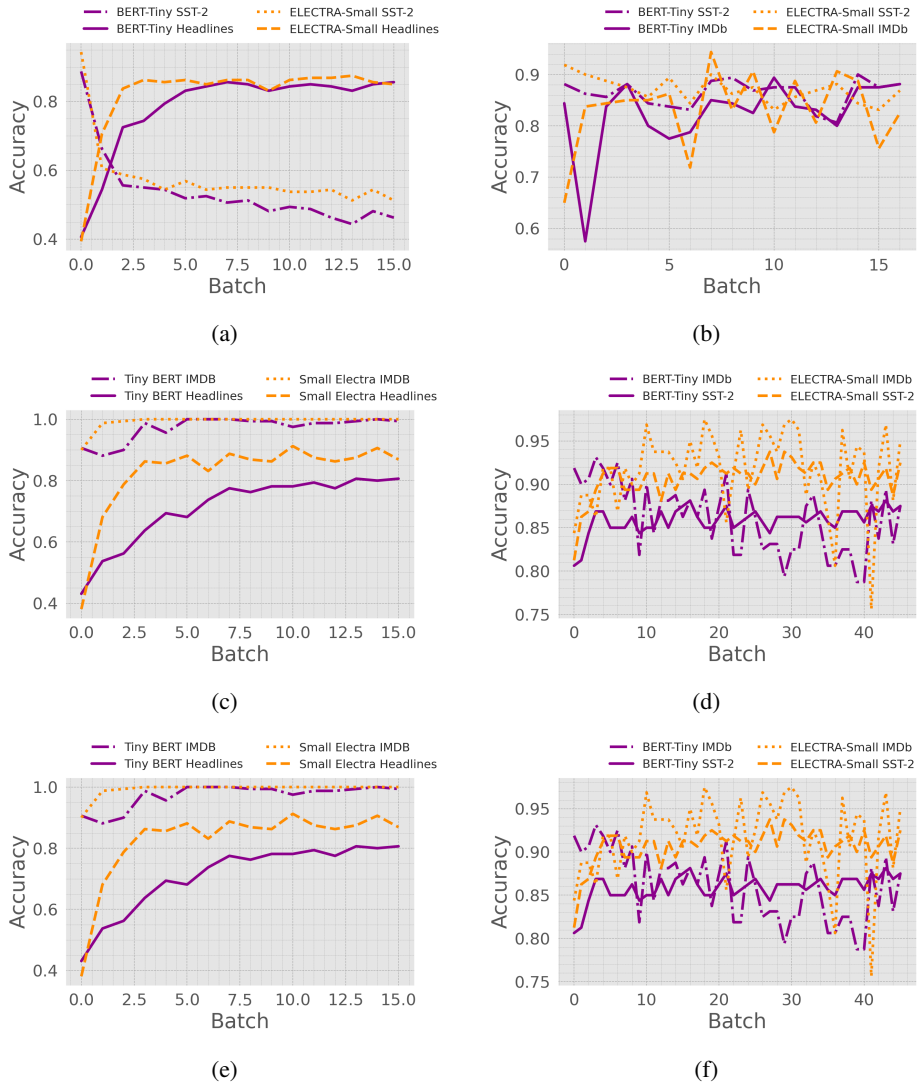


Figure 12: Test set accuracy of BERT-Tiny (purple) and ELECTRA-Small (orange) models while fine-tuning on a secondary dataset after fine-tuning on a primary dataset. The primary and secondary datasets are: (a) Headlines and SST-2; (b) Headlines and IMDb; (c) SST-2 and Headlines; (d) SST-2 and IMDb; (e) IMDb and Headlines; (f) IMDb and SST2. In cases where forgetting happens, BERT-Tiny demonstrates a faster degradation in performance compared to ELECTRA-Small following subsequent fine-tuning.

investigate the impact of random dropout of attention connections on the ORC distribution in *L-ViT* and BERT-Tiny. During training, a Bernoulli mask was implemented to stochastically remove edges across all attention maps at varying rates. In the testing phase, we assess the robustness of the trained models with unmanipulated attention, quantified by $\rho$ as defined in Equation 8, over dataset subsets (see Appendix D.2). Informed by prior research on connection masking techniques (Luo et al., 2021; Zehui et al., 2019), we posit that its application will lead to enhanced model robustness. While the number of active attention connections may decrease, we anticipate this effect will not significantly impair overall performance.

Observing the $\rho$ values at different Bernoulli masking rates on Figure 13, our experiments confirm that this robustness tends to be achieved. Noticeably, a trend toward increasing robustness is often observed when masking rates fall within the range of 0.2 to 0.5, indicating that masking at these rates promotes increased robustness. Furthermore, low performance degradation is observed even at